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Project Title:	Development of an Optical Microcomb System in Support of Fundamental Physics Research on the International Space Station		
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Comments:			
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COI Name (Institution):	Papp, Scott Ph.D. (National Institute of Standards & Technology) Vahala, Kerry Ph.D. (California Institute of Technology)		
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Task Description:	Our research program will investigate and develop advanced microresonator optical frequency comb technology in support of planned European Space Agency (ESA) activities on the International Space Station (ISS). Optical frequency combs, which provide a dense set of absolute frequency reference lines, are an integral component of modern optical atomic clocks. They provide a link between optical and microwave signals and a means to access the stability of optical clocks at any wavelength. For example, a frequency comb will be required for the Space Optical Clock (SOC) experiment to achieve all of its fundamental scientific goals by providing the critical phase coherent optical or microwave link between the atomic frequency standard on the ISS and ground-based clocks. The same frequency comb technology will be a required component of other proposed ESA and NASA fundamental-physics clock experiments, as well as an enabling technology for ESA's planned Quantum Weak Equivalence Principle (QWEP) atom interferometry experiments. Microresonator combs (microcombs) can be lithographically patterned and fabricated on a single silicon chip with dimensions of a few square centimeters and therefore offer revolutionary advantages beyond any other conventional laser comb technology in terms of size, power consumption, and sensitivity to their environment. However, the full capability of a microcomb for precision optical frequency metrology has not been realized. The laboratory experiments we propose will, for the first time, characterize a microcomb's stability with respect to known optical standards, thus verifying its suitability to support the high-performance optical clocks envisioned for SOC and other space missions. Our microcomb system will be tested and interfaced with state-of-the-art optical clocks based on strontium and ytterbium atoms, which are the two candidate SOC platforms. Furthermore, the microcomb will have an output at 1064 nm and a countable microwave repetition rate in the 10-50 GHz range, which are compatible with the optical and microwave links required to achieve the fundamental physics goals of the SOC mission. Our research will develop advanced technology to enable full on-chip integration of a microcomb with essential components for time and frequency metrology including high-Q silica microresonators, clock-signal coupling, comb self-referencing, and fiber v-groove end-couplers. These efforts will contribute to the long-term goal of realizing a single chip-scale device operating at the convergence of optics, frequency metrology, opto-mechanics, and microwave technology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The best optical clocks now keep time with uncertainty measured in the 18th digit; however, their complexity constrains such remarkable precision to a research laboratory setting. Our research is focused on developing key components of optical clocks that will eventually enable their use outside the lab and in outer space. In particular, we aim to develop the compact “clockwork” that will enable the counting of optical cycles on the femtosecond time scale, as well as the synthesis of optical and microwave frequencies critical for transmission and comparison of multiple atomic clocks. Precision time keeping, as explored in our research, is anticipated to impact a wide variety of human activities both in Earth and space. Not only do clocks provide an accurate tool to analyze the inner workings of atoms and molecules, but they are critical to communication and navigation systems that we rely on every day. Examples include the global positioning system and cellular phone networks. Our research is an important step in developing the next generation clocks which will not only be more accurate, but also more compact and easily fabricated.
Task Progress:	In this period of the program, our research activities were focused in two tasks: 1. On-demand generation and stabilization of single soliton pulses in silica microresonators 2. Generation of octave span supercontinuum in silicon nitride and self-referencing of microresonator frequency combs solitons. Task 1: On-demand generation and stabilization of single solitons in silica microresonators. Solitons in optical microresonators have provided a natural mode-locking mechanism for Kerr combs. However, this regime has proven elusive due to thermal effects that shift the cavity resonance as a result of the intensity-dependent refractive index. In general, the microcavity is thermally stable when the pump laser is blue-detuned from resonance but unstable for a red-detuned pump, giving rise to a thermal bistability. Further, in devices with microwave repetition rates, the detuning range over which the soliton exists is small, necessitating active feedback on the detuning. We have developed a new way of generating solitons that circumvents the thermal bistability by sweeping the pump laser over resonance on a timescale faster than the characteristic thermal times of the device. Because this method avoids long thermal triangles, the negative effects of mode crossings are also alleviated, allowing us to generate solitons over a wide range of pump frequencies. For these reasons, our method has numerous advantages over existing schemes and can be considered a universal technique for accessing the soliton regime in Kerr combs. For experiments, we use a silica wedge resonator with FSR ~ 22 GHz, Q ~ 200 million. The resonator sits atop a silicon substrate, and light is coupled into the resonator via a tapered fiber. A two-step process is implemented for the generation and stabilization of solitons. First, a fast and precise scan over the resonance extends the soliton step to hundreds of microseconds thanks to the reduced effects of thermal transients. Second, an offset Pound Drever Hall (PDH) lock is turned on while in the soliton state in order to precisely lock the detuning. Once the lock is turned on, the pump laser can “chase” the resonance through the remaining thermal transients, maintaining a detuning set by the modulation frequency used in the PDH lock. The fast frequency scans of the pump laser are done using a Single Side Band (SSB) modulator with a suppressed carrier. A voltage controlled oscillator (VCO) controls the modulation frequency and therefore the pump frequency, and has arbitrary speed and range compared to the speed and magnitude of thermal effects in the resonator. The bandwidth of the VCO is also useful for feedback to the pump frequency, as bandwidths of >100 kHz can be used in the detuning lock. There are numerous advantages of the SSB-SC technique. First, solitons are stabilized despite the obvious presence of mode crossings. This allows a wide range of pump frequencies to be used, useful for many frequency-comb applications. Second, high pump powers many factors above threshold are not required. Other methods of soliton generation, such as slow frequency scans and power kicks, rely on high optical powers (even if only for short periods of time) to overcome thermal transients induced by the transition from chaos into solitons. Finally, the detuning can be controlled exactly through the phase modulation frequency, and this is often desirable because many soliton pulse parameters (such as duration, repetition rate, and power) are detuning dependent. Task 2: Generation of octave supercontinuum in silicon nitride and self-referencing of microresonator frequency comb solitons. Frequency comb sources with pulse repetition rates larger than 10 GHz, such as microresonator combs, have been challenging to spectrally broaden due to their low pulse energies. Nevertheless, octave-span supercontinuum is

required for full stabilization of a microcomb. Recently, we have used silicon nitride (SiN) waveguides to produce spectra having optical bandwidth beyond an octave. Furthermore, we demonstrate that spectral coherence is maintained in the supercontinuum process by detecting the comb offset frequency at the second-harmonic of the 1550 nm pump laser.

The waveguide devices used in this work are rectangular silicon nitride waveguides with an oxide-clad geometry and a length of 15 mm, thickness of 800 nm and width of 1800 nm. To design a suitable waveguide for self-referencing and ultimately for envisioned clock experiments, numerical simulations of the pulse propagation and subsequent spectral broadening are performed using the generalized nonlinear Schroedinger equation (NLSE) as part of the PyNLO software package. Included in these calculations is a chromatic dispersion profile for each waveguide geometry obtained using a finite difference mode solver implemented in the EMpy software package. We observe precise agreement between our numerical designs and our generated experimental supercontinuua. In this case, the experimental spectra were generated with the output from a more conventional mode-locked Er:fiber laser.

Once characterized, these SiN waveguides are employed for self-referencing 10+ GHz frequency comb sources. In all cases, a pulse shaper is first used to compensate fiber-path dispersion to achieve a compressed pulse at the output of a 4 W amplifier. After amplification, the pulse undergoes normal-dispersion broadening in 5 m of highly-nonlinear fiber before final compression with a grating pair. This compressed pulse is then free-space coupled into the waveguide with an insertion loss of approximately -1.5 dB. Using this configuration, both a 30 GHz EO (electro optic) comb source and a 15 GHz microcomb source have been used to generate coherent supercontinuum. In one case, the EO comb seeds the waveguide with 80 pJ, 100 fs pulses. The SC spectrum exhibits a high-degree of smoothness between 800 nm and 1300 nm. Dispersive-wave enhancement of the spectrum at 775 nm can be used to detect the comb offset frequency. However, due to the multiplication of the noise in the source RF synthesizer, the linewidth is >10 MHz and the signal-to-noise ratio is limited to about 20 dB.

Alternatively, the waveguides can be pumped with a 15 GHz silica-disc soliton microresonator source (as described in Task 1). As with the EO comb, offset-frequency detection is enabled by efficient dispersive wave generation. Significantly, the linewidth of the microcomb offset frequency is sub-MHz. Full stabilization of the comb has been accomplished by locking both the repetition rate, using pump laser feedback, and the comb offset, using an acousto-optic modulator. With the significantly increased spectral coverage shown here, these developing comb technologies are now positioned to provide real benefits to compact optical clocks as well as other frequency comb applications.

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